9 HEAT STRUCTURES

These are used in NEW and RESTART type problems and are required only if heat structures are described. The heat structure card numbers are of the form 1CCCGXNN where the fields are defined as follows:

CCC is a heat structure number. The heat structure numbers need not be consecutive. We suggest, but the system does not require, that if heat structures and hydrodynamic volumes are related, they be given the same number.

G is a geometry number. The combination CCCG is a heat structure geometry combination referenced in the heat structure input data. The G digit is provided to differentiate between different types of heat structures (such as fuel pins and core barrel) that might be associated with the same hydrodynamic volume.

X is the card type.

NN is the card number within a card type.

9.1 General Heat Structure Data, 1CCCG000

This card is required for heat structures. Use 8 words for new data input or 1 word for deleting a heat structure.

W1(I) Number of axial heat structures with this geometry, nh. This number must be > 0 and < 100.

W2(I) Number of radial mesh points for this geometry, np. This number must be < 100. Enter > 1 if no reflood is specified, and > 2 if reflood or metal water reaction is specified.

W3(I) Geometry type. Enter 1 for rectangular, 2 for cylindrical, and 3 for spherical. Spherical geometry is not allowed if reflood is specified. Cylindrical geometry must be specified when the gap conductance model is used.

W4(I) Steady-state initialization flag. Use 0 if the desired initial condition temperatures are entered on input cards 1CCCG401-G499; use 1 if the steady-state initial condition temperatures are to be calculated by the code. If option 1 is chosen, the user is still required to enter temperatures on cards 1CCCG402-G499. In this case, the temperatures are used as starting points for the steady-state solutions. The user should therefore enter temperatures either below or above the minimum film boiling point to assure the respective pre-DNB or post-DNB steady-state condition is calculated. This is because the boiling curve is multi-valued.

W5(R) Left boundary coordinate (m, ft).

W6(I) Reflood condition flag. This quantity is optional if no reflood calculation is to be performed. This quantity may be 0, 1, 2, or a trip number.

If 0, no reflood calculation is to be performed.

If nonzero, all the heat structures in this heat structure/geometry are assumed to form a two-dimensional representation of a fuel pin. The radial mesh is defined on card 1CCCG1NN. Each heat structure represents an axial level of the fuel pin, with the first heat structure being the bottom level. Each heat structure should be connected to a hydrodynamic volume representing the same axial section of the coolant channel. The length of the axial mesh in the fuel pin is given by the height of the connected hydrodynamic volume. If the heat structure is fuel pins or heat exchanger tubes, the length factor (Word 5 on cards 1CCCG501-G509) is the product of the hydrodynamic volume length and the number of pins or tubes. The heat structures represent the temperatures at the midpoint of the axial mesh. Once the reflood calculation is initiated, additional mesh lines are introduced at each end of the fuel pin and between the heat structures. Once the reflood calculation is initiated, it remains activated, and the two-dimensional heat conduction calculation uses a minimum of $2 \bullet nh + 1$ axial mesh nodes. Additional mesh lines are introduced and later eliminated as needed to follow the quench front.

If 1 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa, and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.9 (i.e., nearly empty).

If 2 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.1 (i.e., dryout begins).

If a trip number is entered, the reflood calculation is initiated when the trip is set true. When using the expanded trip number format, 1 and 2 are possible trip numbers. A 1 or 2 entered in this word is not treated as a trip number.

W7(I) Boundary volume indicator. Enter 0 or 1 to indicate that reflood heat transfer applies to the left or right boundary, respectively.

W8(I) Maximum number of axial intervals. Enter 2, 4, 8, 16, 32, 64, or 128 to indicate the maximum number of axial subdivisions a heat structure can have. Storage is allocated for the number indicated, even though a transient may not require that level of subdivision.

9.2 Heat Structure Deletion, 1CCCG000

This card can only be entered for RESTART problems. If entered, all heat structures associated with the heat structure geometry number CCCG are deleted.

W1(A) Enter DELETE.

9.3 Gap Data, 1CCCG001

This card is optional and is needed only if the gap conductance model is to be used. If the card is entered, Word 1 of card 1CCCG100 must be 0, cards 1CCCG011-G099, and cards 1CCCG201-G299 are required. Word 2 of card 201MMM00 must be 3, and a table of the gas component name and mole fraction must be specified in the gap material data (cards 201MMM01-49).

W1(R) Initial gap internal pressure (Pa, lb_f/in^2).

W2(I) Gap conductance reference volume. This word is required. The pressure of the gas in a fuel pin for the gap conductance model is given by $P(t) = P(0)/T(0) \bullet T(t)$, where P(t) is the pressure in the fuel pin and T(t) is the temperature in the reference volume. P(0) is Word 1 above, and T(0) is the initial value, if the volume is also being defined with these input data or the value from the restart block. The reference volume is usually the hydrodynamic volume (i.e., the nine-digit number CCCNN0000) most closely associated with the nonfuel region in a fuel pin at the top of a stack of fuel pellets.

W3(R) Gap conductance multiplier. This word is optional. The default value is 1.0.

9.4 Metal-Water Reaction Control, 1CCCG003

This card is optional, and if this card is not present, no metal-water reaction will be calculated. The initial oxide thickness is assumed to be 0 on the inner surface. It remains 0 unless cladding rupture occurs. CCCG is a heat structure geometry number.

W1(R) Initial oxide thickness on cladding's outer surface (m, ft).

9.5 Fuel Cladding Deformation Model Control, 1CCCG004

If this card is optional, and if it is not present, no deformation calculations will be done. If this card is present, then card 1CCCG001 must also be present. CCCG is a heat structure geometry number.

W1(I) Form loss factor flag. Enter 0 if no additional form loss factors are to be calculated after a rod ruptures. Enter 1 if additional form loss factors are to be calculated. Either a 0 or a 1 must be entered.

9.6 Gap Deformation Data, 1CCCG011-G099

These cards are required for the gap conductance model only. The card format is sequential format, five words per set, describing nh heat structures.

W1(R) Fuel surface roughness (m, ft). This number must be ≥ 0. An appropriate value is 10^{-6} m. A negative entry is reset to 10^{-6} m with no errors.

W2(R) Cladding surface roughness (m, ft). This number must be either positive or 0. An appropriate value is 2 x 10^{-6} m. A negative entry is reset to 2 x 10^{-6} m with no errors.

W3(R) Radial displacement due to fission gas-induced fuel swelling and densification (m, ft). This number must be ≥ 0. A negative entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.

W4(R) Radial displacement due to cladding creepdown (m, ft). The value is normally negative. A positive entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.

W5(I) Heat structure number.

9.7 Heat Structure Mesh Flags, 1CCCG100

This card is required for heat structure input.

W1(I) Mesh location flag. If 0, geometry data, including mesh interval data, composition data, and source distribution data, are entered with this heat structure input. If nonzero, that information is taken from the geometry data from the heat structure geometry (CCCG) number in this word. If this word is nonzero, the remaining geometry information described in Section 9.7-Section 9.10 is not entered.

W2(I) Mesh format flag. This word is needed only if Word 1 is 0, though no error occurs if it is present when Word 1 is nonzero. The mesh interval data are given as a sequence of pairs of numbers in one of two formats to be used in cards 1CCCG101-G199. If this word is 1 (format 1 on cards 1CCCG101-G199), the pairs of numbers contain the number of intervals in this region and the right boundary coordinate. For the first pair, the left coordinate of the region is the left boundary coordinate previously entered in Word 5 of card 1CCCG000; for succeeding pairs, the left coordinate is the right coordinate of the previous pair. If this word is 2 (format 2 on cards 1CCCG101-G199), the format is a sequential expansion of mesh intervals; i.e., the distance in Word 1 on cards 1CCCG101-G199 is used for each interval starting from the leftmost, as yet unspecified, interval to and including the interval number specified in Word 2.

9.8 Format 1 Number of Mesh Intervals, 1CCCG101-G199

These cards are required if Word 1 of card 1CCCG100 is 0. The sum of the numbers of intervals must be np-1. The card numbers need not be sequential.

W1(I) Number of intervals. Enter the number of intervals, not the interval number.

W2(R) Right coordinate (m, ft).

9.9 Format 2 Mesh Intervals, 1CCCG101-G199

These cards are required if Word 1 of card 1CCCG100 is 0. The sequential expansion must be for np-1 intervals. The card numbers need not be sequential.

W1(R) Mesh interval (m, ft.)

W2(I) Interval number.

9.10 Heat Structure Composition Data, 1CCCG201-G299

These cards are required if Word 1 of card 1CCCG100 is 0 and must not be entered otherwise. The card format is two numbers per set in sequential expansion format for np-1 intervals. The card numbers need not be in sequential order.

W1(I) Composition number. The absolute value of this quantity is the composition number, and it must be identical to the subfield MMM used in Heat Structure Thermal Property Data, Section 10. The sign indicates whether the region over which this composition is applied is to be included or excluded from the volume averaged temperature computation. If positive, the region is included; if negative, the region is not included. The option to exclude regions from the volume averaged temperature integration is to limit the integration to fuel regions only for use in reactivity feedback calculations. Gap and cladding regions should not be included in this case. If the gap conductance model is used, only one interval can be used for the gap model.

W2(I) Interval number.

9.11 Fission Product Decay Heat Flag, 1CCCG300

This card is required if fission product decay heat is used on this heat structure. This card sets the fission product decay heat flag. The code will then treat card 1CCCG301 as a gamma attenuation coefficient card. This card is not needed if fission product decay heat is not used on this heat structure.

W1(A) DKHEAT.

9.12 Heat Structure Source Distribution Data, 1CCCG301-G399

These cards are required if Word 1 of card 1CCCG100 is 0 and must not be entered otherwise. The card format is two numbers per set in sequential expansion format for np-1 intervals. The card numbers need not be in sequential order. Radial power peaking factors are entered here.

W1(R) Source value. These are relative values only and can be scaled by any factor without changing the results. By entering different values for the various mesh intervals, a characteristic shape of a power curve can be described.

W2(I) Mesh interval number.

If card 1CCCG300 is entered, then card 1CCCG301 is treated as a gamma attenuation coefficient card.

W1(R) Gamma attenuation coefficient. These are values dependent on the heat structure material. A value of 50 is recommended for stainless steel.

W2(I) Mesh interval number.

9.13 Initial Temperature Flag, 1CCCG400

This card is optional; if missing, Word 1 is assumed to be 0.

W1(I) Initial temperature flag. If this word is 0 or -1, initial temperatures are entered with the input data for this heat structure geometry. If greater than 0, initial temperatures for this heat structure geometry are taken from the heat structure geometry number in this word, and the initial temperature distribution from Section 9.14 or Section 9.15 is not needed.

9.14 Format 1 Initial Temperature Data, 1CCCG401-99

These cards are required if Word 1 of card 1CCCG400 is 0. One temperature distribution is entered; and the same distribution is applied to all of the nh heat structures. The card format is two numbers per set in sequential expansion format for np mesh points

W1(R) Temperature (K, ^{o}F).

W2(I) Mesh point number.

9.15 Format 2 Initial Temperature Data, 1CCCG401-499

These cards are required if Word 1 of card 1CCCG400 is -1. A separate temperature distribution must be entered for each of the nh heat structures. The distribution for the first heat structure is entered on card 1CCCG401, the distribution for the second heat structure is entered on card 1CCCG402, and the remaining distributions are entered on consecutive card numbers. Continuation cards can be used if the data do not fit on one card.

W1-WNP(R) Temperature (K, ^{o}F). Enter the np mesh point temperatures in order from left to right.

9.16 Left Boundary Condition, 1CCCG501-599

These cards are required. The boundary condition data for the heat structures with this geometry are entered in a slightly modified form of sequential expansion using six quantities per set for the number of heat structures with this geometry (nh sets). The modification deals with Words 1 and 2.

W1(I) Boundary volume number or general table. This word specifies the hydrodynamic volume number (of the form CCCNN000F) or general table associated with the left surface of this heat structure. These are used to specify the sink temperature. If 0, no volume or general table is associated with the left surface of this heat structure, and a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient at the boundary), or a temperature of 0 is used for a surface temperature or a sink temperature in boundary conditions. A boundary volume number is entered as a positive number. If F is 0 or 4, the volume coordinate associated values such as average volume velocity are taken from the x-coordinate; if F is 2 or 1, volume coordinate associated values are taken from the y- or z-axes, respectively. Specifying a volume coordinate not in use is an input error. A general table is entered as a negative number (-1 through -999).If F is 7, the 3D hydraulic volume is used. Then ccc is a channel number and nn is a mesh number

W2(I) Increment. This word and Word 1 of this card are treated differently from the standard sequential expansion. Word 1 of the first set applies to the first heat structure of the heat structure geometry set. The increment (normally 10000) is added to Word 1, which results in the hydraulic cell number associated with heat structure 2; etc. The increment is applied up to the limit in Word 6 of a set. Word 1 of the next set applies to the next heat structure, and increments are applied as for the first set. The increment may be 0 or nonzero, positive or negative. If Word 1 is 0, this word should be 0.

W3(I) Boundary condition type.

If 0, a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient is used at the boundary). The boundary volume must be 0.

If 1 or 1nn, a convective boundary condition where the heat transfer coefficient obtained from Heat Transfer Package 1 is used. The allowed values of nn are shown in **Table 9.16-1**. The sink temperature is the temperature of the boundary volume. Word 1 must specify a boundary volume with this boundary condition type. The boundary volume cannot be a time-dependent volume.

There are several numbers allowed for Word 3 to activate convective boundary conditions for nonstandard geometries. A 1, 100, or 101 give the default values. The numbers 1, 100, and 101 use the same correlations. The number 101 is recommended; the numbers 1 and 100 are allowed so that the code is backwards compatible with previous input decks. The default convection and boiling correlations were derived mainly based on data from internal vertical pipe flow. Other possible input values are shown in **Table 9.16-1**. When modeling a vertical bundle, the rod or tube pitch-to-diameter ratio should be input on the 801 card. This has the effect of increasing the convective part of heat transfer such that users can input the true hydraulic diameter and get reasonable predictions.

Table 9.16-1 Card 501 Word 3 Convection Boundary Type

Word 3	Geometry Type
1, 100, 101	Default
102	Parallel plates (ORNL, ANS reactor; set gap and span on CCC3101-3199 hydro cards for pipes and CCC0111 hydro card for single volumes and branches, set b = 2 in volume control flag on CCC1001-1099 hydro cards for pipes and CCC0101-0109 hydro cards for single volumes and branches).
109	HANARO fuel
110	Vertical bundle without crossflow (set P/D on 801 card)
111	Vertical bundle with crossflow (set P/D on 801 card)
114	Helical S/G tube side (set Dc/di on 801/901 card Word 10)
124	CANDU fuel bundle element heat transfer
130	Flat plate above fluid
131	HANARO horizontal plate heat exchanger
134	Horizontal bundle
135	Helical S/G shell side (set P/D on 801/901 card Word 10)
140	Modified Dittus-Boelter model for gas turbulent forced convection
141	Gnielinski model for gas turbulent forced convection
142	Olson model for gas turbulent forced convection
143	Olson model for gas turbulent forced and other low *Re* number heat transfer models with wall heating (recommended for use in core)
144	Olson model for gas turbulent forced convection and other low *Re* number heat transfer models with wall cooling (recommended for use in heat exchangers)

If 1000, the temperature of the boundary volume or the temperature from the general table (as specified in Word 1) is used as the left surface temperature. If Word 1 is 0, the surface temperature is set to 0.

If 1xxx, the temperature in general Table xxx is used as the left surface temperature.

If 2xxx, the heat flux from Table xxx is used as the left boundary condition.

If 3xxx, a convective boundary condition is used where the heat transfer coefficient as a function of time is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If 4xxx, a convective boundary condition is used where the heat transfer coefficient as a function of surface temperature is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If reflood is specified, the left boundary condition type must be same for all nh heat structures and, similarly, for the right boundary condition type. The left and right boundary types need not be the same, but neither can be 1000 or 1xxx.

W4(I) Surface area code. If 0, Word 5 is the left surface area. If 1, Word 5 is (a) the surface area in rectangular geometry, (b) the cylinder height or equivalent in cylindrical geometry, or (c) the fraction of a sphere (0.5 is a hemisphere) in spherical geometry.

W5(R) Surface area or factor. As indicated in Word 4, this word contains the surface area (m^2, ft^2) or a geometry dependent multiplier (m^2, ft^2) for rectangular; (m, ft) for cylindrical; or dimensionless for spherical geometries). If the symmetry boundary condition is specified (Word 3 = 0), this word must still be entered nonzero.

W6(I) Heat structure number.

9.17 Right Boundary Condition, 1CCCG601-G699

These cards are required. The left and right surface areas must be compatible with the geometry. The boundary condition data for the heat structures with this geometry are entered in a slightly modified form of sequential expansion using six quantities per set for the number of heat structures with this geometry (nh sets). The modification deals with Words 1 and 2.

W1(I) Boundary volume number or general table. This word specifies the hydrodynamic volume number (of the form CCCNN000F) or general table associated with the right surface of this heat structure. These are used to specify the sink temperature. If 0, no volume or general table is associated with the right surface of this heat structure, and a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient at the boundary), or a temperature of 0 is used for a surface temperature or a sink temperature in boundary conditions. A boundary volume number is entered as a positive number. If F is 0 or 4, the volume coordinate associated values such as average volume velocity are taken from the x-coordinate; if F is 2 or 1, volume coordinate associated values are taken from the y- or z-axes, respectively. Specifying a volume coordinate not in use is an input error. A general table is entered as a negative number (-1 through -999). If F is 7, the 3D hydraulic volume is used. Then ccc is a channel number and nn is a mesh number

W2(I) Increment. This word and Word 1 of this card are treated differently from the standard sequential expansion. Word 1 of the first set applies to the first heat structure of the heat structure geometry set. The increment (normally 10000) is added to Word 1, which results in the hydraulic cell number associated with heat structure 2; etc. The increment is applied up to the limit in Word 6 of a set. Word 1 of the next set applies to the next heat structure, and increments are applied as for the first set. The increment may be 0 or nonzero, positive or negative. If Word 1 is 0, this word should be 0.

W3(I) Boundary condition type.

If 0, a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient is used at the boundary). The boundary volume must be 0.

If 1 or 1nn, a convective boundary condition where the heat transfer coefficient obtained from Heat Transfer Package 1 is used. The allowed values of nn are shown in **Table 9.17-1**. The sink temperature is the temperature of the boundary volume. Word 1 must specify a boundary volume with this boundary condition type. The boundary volume cannot be a time-dependent volume.

There are several numbers allowed for Word 3 to activate convective boundary conditions for nonstandard geometries. A 1, 100, or 101 give the default values. The numbers 1, 100, and 101 use the same correlations. The number 101 is recommended; the numbers 1 and 100 are allowed so that the code is backwards compatible with previous input decks. The default convection and boiling correlations were derived mainly based on data from internal vertical pipe flow. Other possible input values are shown in **Table 9.17-1**. When modeling a vertical bundle, the rod or tube pitch-to-diameter ratio should be input on the 901 card. This has the effect of increasing the convective part of heat transfer such that users can input the true hydraulic diameter and get reasonable predictions.

Table 9.17-1 Card 601 Word 3 Convection Boundary Type

Word 3	Geometry Type
1, 100, 101	Default

Table 9.17-1 Card 601 Word 3 Convection Boundary Type

Word 3	Geometry Type
102	Parallel plates (ORNL, ANS reactor; set gap and span on CCC3101-3199 hydro cards for pipes and CCC0111 hydro card for single volumes and branches, set b = 2 in volume control flag on CCC1001-1099 hydro cards for pipes and CCC0101-0109 hydro cards for single volumes and branches).
109	HANARO fuel
110	Vertical bundle without crossflow (set P/D on 901 card)
111	Vertical bundle with crossflow (set P/D on 901 card)
114	Helical S/G tube side (set Dc/di on 801/901 card Word 10)
124	CANDU fuel bundle element heat transfer
130	Flat plate above fluid
131	HANARO horizontal plate heat exchanger
134	Horizontal bundle
135	Helical S/G shell side (set P/D on 801/901 card Word 10)
140	Modified Dittus-Boelter model for gas turbulent forced convection
141	Gnielinski model for gas turbulent forced convection
142	Olson model for gas turbulent forced convection
143	Olson model for gas turbulent forced and other low *Re* number heat transfer models with wall heating (recommended for use in core)
144	Olson model for gas turbulent forced convection and other low *Re* number heat transfer models with wall cooling (recommended for use in heat exchangers)

If 1000, the temperature of the boundary volume or the temperature from the general table (as specified in Word 1) is used as the right surface temperature. If Word 1 is 0, the surface temperature is set to 0.

If 1xxx, the temperature in general Table xxx is used as the right surface temperature.

If 2xxx, the heat flux from Table xxx is used as the right boundary condition.

If 3xxx, a convective boundary condition is used where the heat transfer coefficient as a function of time is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If 4xxx, a convective boundary condition is used where the heat transfer coefficient as a function of surface temperature is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If reflood is specified, the right boundary condition type must be same for all nh heat structures and, similarly, for the left boundary condition type. The left and right boundary types need not be the same, but neither can be 1000 or 1xxx.

W4(I) Surface area code. If 0, Word 5 is the right surface area. If 1, Word 5 is (a) the surface area in rectangular geometry, (b) the cylinder height or equivalent in cylindrical geometry, or (c) the fraction of a sphere (0.5 is a hemisphere) in spherical geometry.

W5(R) Surface area or factor. As indicated in Word 4, this word contains the surface area (m^2, ft^2) or a geometry dependent multiplier (m^2, ft^2) for rectangular; (m, ft) for cylindrical; or dimensionless for spherical geometries). If the symmetry boundary condition is specified (Word 3 = 0), this word must still be entered nonzero.

W6(I) Heat structure number.

9.18 Source Data, 1CCCG701-G799

These cards are required for heat structure data. The card format is sequential expansion format, five words per set, describing nh heat structures.

W1(I) Source type. If 0, no source is used. If a positive number is less than 1000, power from the general table with this number is used as the source. If 1000-1004, the number has the form 100t, and the source is taken from a point kinetics calculation. The field t = 0 specifies total reactor power, t = 1 specifies total decay power, t = 2 specifies fission power, t = 3 specifies fission product decay power, and t = 4 specifies actinide decay power. If 10001-19999, the source is the control variable whose number is this quantity minus 10000. If -3 is entered, the source of the heat structure is taken from the 3D kinetics module, MASTER. In this case, Input Cards 61100001 and 61100002 should be provided. In addition, MASTER input files and mapping data ("MAS_MAP" file) should be also prepared [KAERI/TR-2232/2002].

W2(R) Internal source multiplier. Axial peaking factors may be entered here. These values are multiplied by the power in the general table number in Word 1 to obtain the total power generated in this heat structure. These factors are not relative factors.

W3(R) Direct moderator heating multiplier for left boundary volume.

W4(R) Direct moderator heating multiplier for right boundary volume.

W5(I) Heat structure number.

9.19 Additional Left Boundary Option, 1CCCG800

W1(I) If this card is not entered or if this word is 0, the nine-word format is used on cards 1CCCG801-G899. If this word is 1, the twelve-word format is used on the cards. If this word is 2, the thirteen-word format is used on the cards (needed for PG-CHF correlation). If this word is 3, the nine-word format is used on the cards and model multiplier inputs are required. If this word is 4, the twelve-word format is used on the cards and model multiplier inputs are required. (see MARS084)

W2(R) Multiplier for liquid heat transfer mode (default = 1.0)

W3(R) Multiplier for nucleate boiling mode (default = 1.0)

W4(R) Multiplier for AECL CHF value (default = 1.0)

W5(R) Multiplier for transition boiling mode (default = 1.0)

W6(R) Multiplier for film boiling mode (default = 1.0)

W7(R) Multiplier for vapor heat transfer mode (default = 1.0)

W8(I) Trip number to activate multipliers. The 6 multipliers are applied only during the period when trip is true.

9.20 9-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 0 on card 1CCCG800. The cards are in sequential expansion format, nine words per set, describing nh heat structures. Sequential expansion would only be used where the critical heat flux value was not of importance, since the length to all heat structures in the expansion would be the same. Words 2-8 are used for the CHF correlation.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since (heated perimeter) $\leq$ (wetted perimeter). It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.

W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.

W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.

W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.

W9(I) Heat structure number.

9.21 12-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 1 on card 1CCCG800. The cards are in sequential expansion format, twelve words per set, describing nh heat structures.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since (heated perimeter) ≤ (wetted perimeter). It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.

W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.

W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.

W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

W10(R) Rod or tube pitch-to-diameter ratio (P/D). The default is 1.1. The maximum is 1.6. It is not used unless Word 3 on the 501 card is 110, 111, 114, 124 or 135. If Helical geometry (114 option on 501 CARD)was selected, this value is Helical Circle Diameter-to-Tube Inner Diameter Ratio (Dc/di), and unlimited maximum value can be entered

W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.

W12(I) Heat structure number.

9.22 13-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 2 on card 1CCCG800. The cards are in sequential expansion format, thirteen words per set, describing nh heat structures.

W1(R) Not used, enter 0.0.

W2(R) Reduced heated length forward (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel inlet to the point of the predicted CHFR when the liquid volume velocity is positive or 0. The second term is the ratio of average heat flux from the heated channel inlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y. Word 2 should be determined as follows:

$$y \bullet T_a = \frac{1.}{q(y)}\int_0^y q(z)dz.$$

W3(R) Reduced heated length reverse (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel outlet to the point of the predicted CHFR when the liquid volume velocity is negative. The second term is the ratio of average heat flux from the heated channel outlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y. Word 3 should be determined as follows:

$$y \bullet T_a = \frac{1.}{q(y)}\int_0^y q(z)dz.$$

W4(R) Grid spacer factor forward. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 4 must be input as W4 = 1.0.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 4 should be input either as W4 = 1.0 / $\overline{R}$, if the statistical evaluation data for the rod bundles are available ($\overline{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as W4 = 1.0, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 4 should be input as: W4 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W4 should be input as W4 = 1.0.

W5(R) Grid spacer factor reverse. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 5 must be input as W5 = 1.0.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 5 should be input either as W5 = 1.0 / $\overline{R}$, if the statistical evaluation data for the rod bundles are available ($\overline{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as W5 = 1.0, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 5 should be input as: W5 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W5 should be input as W5 = 1.0.

W6(R) Factor of the radial heat flux distribution. This should be input as:

$$T_r = q\frac{\sum_i r_i}{\sum_i r_i q_i}$$

This is the ratio of local heat flux on referred perimeter to average heat flux on perimeters pertaining to the subchannel (or the annulus).

W7(I) Heated channel upstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the inlet for the heated channel. This is to get the heated channel inlet quality in the case of forward flow direction.

W8(I) Heated channel downstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the outlet for the heated channel. It applies when the flow reverses.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 501 card is 110, 111, 114, or 135. If Helical geometry (114 option on 501 CARD)was selected, this value is Helical Circle Diameter-to-Tube Inner Diameter Ratio (Dc/di), and unlimited maximum value can be entered. If CANDU geometry (124 option on Card 1CCCG5XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0

W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.

W12(I) CHFR correlation option. This is input in mn format. The first digit specifies the CHFR correlation form.

If m = 1, then the basic form of PG CHFR correlation is used.

If m = 2, then the flux form of the PG CHFR correlation is used.

If m = 3, then the geometry form of PG CHFR correlation is used.

If m = 4, then the power form of PG CHFR correlation is used.

The second digit specifies the geometry of heated structure. If this is the rod bundle, it specifies if and how the statistical evaluation data are applied for the grid spacer factor (see Word 4 and Word 5).

If n = 1, then this is the tube.

If n = 2, then this is the internally heated annulus.

If n = 3, then this is the rod bundle. The use of an isolated subchannel model is recommended. This is used if the applicable PG CHFR correlation statistical evaluation data are not available.

If n = 4, then this is the rod bundle. The use of an isolated subchannel model is recommended. An extended use of the PG CHFR statistical evaluation data is enabled.

If n = 5, then this is the rod bundle. This is only used in combination with m = 1. Applicable for a subchannel code respecting lateral mixing.

W13(I) Heat structure number.

9.23 Additional Right Boundary Option, 1CCCG900

This card is optional, and if it is not entered, the nine-word format is used on cards 1CCCG901-G999.

W1(I) If this word is 0, the nine-word format is used on cards 1CCCG901-G999. If this word is 1, the twelve-word format is used on the cards. If this word is 2, the thirteen-word format is used on the cards (needed for PG-CHF correlation). If this word is 3, the nine-word format is used on the cards and model multiplier inputs are required. If this word is 4, the twelve-word format is used on the cards and model multiplier inputs are required.

W2(R) Multiplier for liquid heat transfer mode (default = 1.0)

W3(R) Multiplier for nucleate boiling mode (default = 1.0)

W4(R) Multiplier for AECL CHF value (default = 1.0)

W5(R) Multiplier for transition boiling mode (default = 1.0)

W6(R) Multiplier for film boiling mode (default = 1.0)

W7(R) Multiplier for vapor heat transfer mode (default = 1.0)

W8(I) Trip number to activate multipliers. The 6 multipliers are applied only during the period when trip is true.

9.24 9-Word Format Additional Right Boundary1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 0 on card 1CCCG900. The cards are in sequential expansion format, nine words per set, describing nh heat structures. Sequential expansion would only be used where the critical heat flux value was not of importance, since the length to all heat structures in the expansion would be the same. Words 2-8 are used for the CHF correlation.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since (heated perimeter) $\leq$ (wetted perimeter). It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.

W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.

W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.

W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.

W9(I) Heat structure number.

9.25 12-Word Format Additional Right Boundary, 1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 1 on card 1CCCG900. The cards are in sequential expansion format, twelve words per set, describing nh heat structures.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since (heated perimeter) $\leq$ (wetted perimeter). It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0. To ignore the length effect, put in a large number (i.e., $\geq$ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.

W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.

W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.

W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

W10(R) Rod or tube pitch-to-diameter ratio (P/D). The default is 1.1. The maximum is 1.6. It is not used unless Word 3 on the 501 card is 110, 111, 114 or 135. If Helical geometry (114 option on 501 CARD)was selected, this value is Helical Circle Diameter-to-Tube Inner Diameter Ratio (Dc/di), and unlimited maximum value can be entered. If CANDU geometry (124 option on Card 1CCCG5XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0

W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.

W12(I) Heat structure number.

9.26 13-Word Format Additional Right Boundary, 1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 2 on card 1CCCG900. The cards are in sequential expansion format, thirteen words per set, describing nh heat structures.

W1(R) Not used, enter 0.0.

W2(R) Reduced heated length forward (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel inlet to the point of the predicted CHFR when the liquid volume velocity is positive or 0. The second term is the ratio of average heat flux from the heated channel inlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y. Word 2 should be determined as follows:

$$y \bullet T_a = \frac{1.}{q(y)}\int_0^y q(z)dz\,.$$

W3(R) Reduced heated length reverse (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel outlet to the point of the predicted CHFR when the liquid volume velocity is negative. The second term is the ratio of average heat flux from the heated channel outlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y. Word 3 should be determined as follows:

$$y \bullet T_a = \frac{1.}{q(y)}\int_0^y q(z)dz .$$

W4(R) Grid spacer factor forward. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 4 must be input as W4 = 1.0.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 4 should be input either as W4 = 1.0 / $\overline{R}$, if the statistical evaluation data for the rod bundles are available ($\overline{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as W4 = 1.0, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 4 should be input as: W4 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W4 should be input as W4 = 1.0.

W5(R) Grid spacer factor reverse. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 5 must be input as W5 = 1.0.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 5 should be input either as W5 = 1.0 / $\overline{R}$, if the statistical evaluation data for the rod bundles are available ($\overline{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as W5 = 1.0, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 5 should be input as: W5 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W5 should be input as W5 = 1.0.

W6(R) Factor of the radial heat flux distribution. This should be input as:

$$T_r = q\frac{\sum_i r_i}{\sum_i r_i q_i}$$

This is the ratio of local heat flux on referred perimeter to average heat flux on perimeters pertaining to the subchannel (or the annulus).

W7(I) Heated channel upstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the inlet for the heated channel. This is to get the heated channel inlet quality in the case of forward flow direction.

W8(I) Heated channel downstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the outlet for the heated channel. It applies when the flow reverses.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 601 card is 110, 111, 114, or 135. If CANDU geometry (124 option on Card 1CCCG6XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0

W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.

W12(I) CHFR correlation option. This is input in mn format. The first digit specifies the CHFR correlation form.

If m = 1, then the basic form of PG CHFR correlation is used.

If m = 2, then the flux form of the PG CHFR correlation is used.

If m = 3, then the geometry form of PG CHFR correlation is used.

If m = 4, then the power form of PG CHFR correlation is used.

The second digit specifies the geometry of heated structure. If this is the rod bundle, it specifies if and how the statistical evaluation data are applied for the grid spacer factor (see Word 4 and Word 5).

If n = 1, then this is the tube.

If n = 2, then this is the internally heated annulus.

If n = 3, then this is the rod bundle. The use of an isolated subchannel model is recommended. This is used if the applicable PG CHFR correlation statistical evaluation data are not available.

If n = 4, then this is the rod bundle. The use of an isolated subchannel model is recommended. An extended use of the PG CHFR statistical evaluation data is enabled.

If n = 5, then this is the rod bundle. This is only used in combination with m = 1. Applicable for a subchannel code respecting lateral mixing.

W13(I) Heat structure number.